

HARLAN, IA, USA

WMO: 722097

Lat: 41.584N

Lon: 95.339W

Elev: 1231

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.2	(c) -0.3	(d) -13.0	(e) 2.8	(f) -5.8	(g) -8.3	(h) 3.7	(i) -0.3	(j) 32.8	(k) 25.5	(l) 28.9	(m) 22.3	(n) 5.4	(o) 190	(p) 0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 91.0	(d) 75.2	(e) 89.7	(f) 75.0	(g) 86.5	(h) 73.3	(i) 79.8	(j) 86.8	(k) 77.7	(l) 84.9	(m) 75.9	(n) 83.4	(o) 11.9	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	149.1	84.0	75.2	138.4	82.0	73.2	129.0	80.5	44.2	87.3	42.1	84.8	40.2	83.7	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	24.2	20.7	-15.6	95.1	5.1	1.7	-19.3	96.4	-22.3	97.4	-25.2	98.3	-28.9	99.6		
			-16.0	83.0	5.1	1.9	-19.7	84.4	-22.7	85.5	-25.6	86.6	-29.3	88.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	22.0	24.6	39.2	49.8	62.1	71.3	73.9	70.9	65.4	52.3	38.7	26.0		
	DBStd	20.84	12.20	12.35	12.17	10.08	9.01	5.89	5.78	5.78	7.83	9.63	10.42	11.34		
	HDD50	3281	867	712	375	124	10	0	0	0	2	87	360	744		
	HDD65	6423	1332	1132	800	463	164	13	7	16	90	408	789	1209		
	CDD50	3220	0	0	41	118	386	640	742	649	464	158	21	1		
	CDD65	887	0	0	2	8	74	203	284	199	102	15	0	0		
	CDH74	8810	0	0	20	140	812	2041	2858	1746	1000	188	5	0		
	CDH80	3013	0	0	3	36	261	733	1082	540	308	50	0	0		
	WSAvg	8.7	9.5	9.5	10.4	11.3	10.3	8.3	5.7	5.4	6.8	8.3	9.4	9.0		
	PrecAvg	33.3	0.9	1.0	2.3	3.5	5.1	4.3	4.4	3.8	3.0	2.2	1.6	1.0		
(15) Precipitation	PrecMax	45.3	1.9	2.0	5.2	7.8	9.5	9.8	8.3	8.5	8.6	6.2	3.3	2.6		
	PrecMin	22.7	0.3	0.1	0.1	0.7	1.5	0.8	1.4	0.4	0.8	0.5	0.1	0.0		
	PrecStd	6.1	0.4	0.6	1.4	1.7	2.5	2.4	2.1	2.1	1.9	1.3	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	61.5	76.7	84.5	92.6	93.3	94.7	91.5	91.0	86.2	72.5	56.7		
		MCWB	42.9	47.5	58.3	62.6	68.1	73.0	77.7	76.2	73.4	68.9	56.8	49.5		
	2%	DB	46.1	53.8	70.3	77.5	86.4	90.3	90.9	89.6	87.9	79.1	65.7	51.6		
		MCWB	40.4	45.2	56.3	58.3	67.8	72.9	77.7	76.6	72.3	63.4	55.3	46.0		
	5%	DB	41.3	45.8	63.8	72.4	81.7	87.8	89.7	86.0	83.6	73.2	60.7	45.8		
		MCWB	37.5	40.5	53.7	56.6	64.7	71.6	77.4	75.6	71.3	59.8	52.0	41.2		
	10%	DB	37.0	41.0	58.7	67.7	78.7	84.1	86.2	82.1	80.7	69.8	54.9	41.2		
		MCWB	34.4	37.4	50.8	55.3	63.5	70.4	75.8	73.2	68.1	58.0	46.9	37.3		
	0.4%	WB	44.9	53.2	62.9	65.6	73.3	79.7	83.3	81.1	77.2	73.0	60.7	53.7		
		MCDB	49.7	60.8	71.6	75.6	83.0	87.0	88.8	88.8	85.6	81.5	64.8	55.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	45.6	59.7	62.1	70.6	76.7	80.7	78.9	75.0	67.9	56.6	45.8		
		MCDB	46.4	51.2	66.4	72.2	80.5	84.6	87.0	86.0	83.3	74.6	64.2	47.6		
	5%	WB	37.7	41.0	55.4	59.8	68.7	74.9	78.5	77.1	73.3	63.4	52.5	41.4		
		MCDB	41.3	45.6	61.8	69.3	78.3	83.5	85.6	83.6	80.5	68.6	59.6	46.0		
	10%	WB	35.1	37.4	50.5	57.0	66.5	72.9	76.6	75.0	71.2	59.3	48.2	37.7		
		MCDB	37.5	41.0	59.7	65.8	74.6	81.3	84.0	81.1	77.7	66.8	55.2	41.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	18.3	20.6	22.8	21.8	20.7	20.6	20.2	23.5	22.9	21.0	17.2		
		MCDBR	22.8	24.7	30.1	31.0	26.5	23.9	22.5	22.8	25.5	29.2	28.8	23.3		
		MCWBR	17.1	17.5	17.3	15.9	11.1	10.6	11.5	12.5	12.1	15.0	17.3	17.7		
	5% WB	MCDBR	21.2	23.8	25.2	24.8	21.4	20.3	19.1	19.8	20.7	21.8	25.9	21.4		
		MCWBR	16.4	17.4	15.0	14.7	10.8	11.7	11.9	12.6	11.5	14.4	17.3	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.310	0.326	0.380	0.403	0.411	0.427	0.413	0.376	0.332	0.295	0.285		
	taud		2.429	2.365	2.396	2.265	2.271	2.305	2.248	2.298	2.357	2.466	2.498	2.440		
	Ebn at Noon		278	285	293	282	277	274	268	269	272	273	271	266		
	Edh at Noon		24	31	34	41	42	41	43	40	35	28	23	23		
(44) All-Sky Solar Radiation	RadAvg		617	930	1200	1525	1821	2017	2047	1760	1458	1004	691	519		
	RadStd		56	56	110	116	136	168	137	121	96	121	49	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page